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Francesco di Giorgio Martini's Treatise on Engineering and Its Plagiarists

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# Francesco di Giorgio Martini's *Treatise on Engineering* and Its Plagiarists

LADISLAO RETI\*

FRANCESCO DI GIORGIO MARTINI (1439-1501), the great Sienese painter, sculptor, and architect, was also interested, as were several of his contemporary fellow craftsmen, in the study and development of mechanical devices. This was in accordance with the still flourishing Vitruvian tradition. His engineering treatise, still little known, is mainly dedicated to civil and military architecture and contains hundreds of small but perfectly drawn illustrations showing war machines of every kind, as well as cranes, mills, pumps, etc. Although a number of studies have been published about the artistic and architectural work of Francesco di Giorgio,<sup>1</sup> his work in technology has only occasionally been noticed.

Francesco's *Trattato di Architectura* is preserved in several original manuscripts: a first draft in the Biblioteca Comunale in Siena (No. S.IV. 4); an elaborated version is in the Biblioteca Nazionale of Florence (No. II.I.141),<sup>2</sup> the presentation copy dedicated to Federico de Montefeltro, Duke of Urbino, is in the former library of the Duke of Genoa (Codice Saluzziano No. 148, parchment). There is also an incomplete manuscript that once belonged to Leonardo da Vinci. This latter is of particular interest because Leonardo added marginal notes and sketches; the manuscript is now in the Laurenziana Library in

\* Dr. Reti of Sao Paulo, Brazil, is an internationally famous chemical engineer. In addition to his technical papers, he has published articles on Leonardo da Vinci.

<sup>1</sup> The most complete bibliography on the subject can be found in Allen Stuart Wellers, *Francesco di Giorgio Martini 1439-1501* (Chicago, Ill., 1943). A specialized bibliography is presented in Roberto Papini, *Francesco di Giorgio Architetto* (3 vols., Florence, 1946), mainly concerned with Francesco's activity as architect. A short and useful monograph is Selwyn Brinton, *Francesco di Giorgio of Siena* (2 vols., London, 1934-35). In his preface Brinton expresses "the hope that at some future day—perhaps not so very far away—this wonderful treatise, which represents the experience and expert judgment of a great Renaissance architect and engineer, may appear in its complete text in English."

<sup>2</sup> Microfilm copy in the Library of Congress (Microfilm M. LA. 588 f.) However, most of the machine drawings are lacking, and others are different from those in the Laurenziano and Saluzziano Codices.

Florence (Codex Mediceo Laurenziano 361, formerly Ashb. 361 [293]). In addition, several old copies of the treatise or its drawings are to be found in other Italian libraries, reflecting the early interest aroused by Francesco's work.

These *Trattato* manuscripts, especially those parts dealing with mechanical engineering and technology, have never been adequately studied or fully published. A fairly accurate picture of Francesco di Giorgio Martini's work was first made available to scholars in 1841 when Carlo Promis, using the Codex owned by Saluzzo, published the *Trattato* for the first time.<sup>3</sup> Unfortunately, Promis transcribed and illustrated only six of the seven books; the seventh, which deals with all kinds of mechanical devices, was omitted except for its remarkable prologue. Hence the Promis edition was of interest only to those concerned with military and civil architecture. Promis's unforgivable omission probably explains why the extremely interesting mechanical drawings of Francesco escaped the notice of historians of technology for the next hundred years.

Promis attempted to justify his omission, inconceivable to a modern scholar, in a footnote (p. 10): "In this last book, although the author promises to deal with mechanics properly, he speaks almost exclusively of the art of mill-making: for this reason it was considered convenient to exclude it from the present edition." Further on (p. 202) Promis reiterates; "(There) . . . follow the drawings and descriptions of fifteen mills, one siphon (pump) moved by bellows and another by levers, with five machines for moving or lifting weights. All these have been left out, being of little importance and because they can be found in all the books of mechanics of the sixteenth century." This was most unfortunate. Because the original manuscripts were scarcely accessible, Promis' statement was taken at its face value and Francesco di Giorgio's book was considered important solely in regard to civil and military architecture. Even as late as 1943, Wellers, describing the *Trattato*, mentions the seventh book only very briefly: "(It) . . . has to do with machinery of architecture—principally weights and pulleys designed to lift building stones."<sup>4</sup>

Further confusion was caused by the fact that the Codex Saluzziano and the Codex Laurenziano, in spite of being written by the same hand and containing almost identical drawings, were, for a long time, not attributed to the same author. Early interest was aroused by the Laurenziana codex because of the marginalia added by Leonardo. Of

<sup>3</sup> Francesco di Giorgio Martini, *Trattato di Architettura Civile e Militare*, edited by Carlo Promis (2 vols., Turin, 1841).

<sup>4</sup> Wellers, *op. cit.*, p. 270.

this the part corresponding to the seventh book of the Saluzzo Codex, omitted by Promis, was at last transcribed by Mancini in 1917.<sup>5</sup> However, the reproduction of the drawings was poor. Furthermore, Mancini emphatically denied that the book *could* have been written by Francesco and, with some hesitation, he attributed authorship to Alberti.

It is not surprising, therefore, that the literature of the history of technology has neglected Francesco di Giorgio's studies of machines and engineering. He is not mentioned in Feldhaus' main work,<sup>6</sup> though in a later book<sup>7</sup> he is remembered as a military engineer. Beck's important work,<sup>8</sup> which discusses and analyzes leading engineers from antiquity to modern times, does not mention Francesco, and Usher<sup>9</sup> gives him but one sentence. Singer's encyclopedic five-volume work<sup>10</sup> ignores him except for a curt reference in the chapter on town-planning. However, Parsons, in his book on Renaissance engineering,<sup>11</sup> comments on a manuscript "which has neither date nor author's name . . . and is a treatise of great value, devoted to architecture, engines of war and mechanical devices of various sorts." His remark that the Codex Laurenziano was at one time in the hands of Leonardo da Vinci identifies it with the Laurenziana Codex 361 already mentioned.

A few pages and drawings from the *Trattato's* appendix have been reproduced and discussed in the books of Canestrini,<sup>12</sup> Forti,<sup>13</sup> and Papini.<sup>14</sup> In Klemm's book<sup>15</sup> Francesco is remembered as one of the 'Künstleringenieure' of the Renaissance. Lynn White's important

<sup>5</sup> *Giorgio Vasari, Vite cinque* (Franceschi, Alberti, Franc. di Giorgio, Signorelli, de Marcillat), annotated by Girolamo Mancini (Florence, 1917).

<sup>6</sup> F. M. Feldhaus, *Die Technik der Vorzeit, der geschichtlichen Zeit und der Naturvölker* (Leipzig and Berlin, 1914).

<sup>7</sup> F. M. Feldhaus, *Die Technik der Antike u. d. Mittelalters* (Potsdam, 1931).

<sup>8</sup> Theodor Beck, *Beiträge zur Geschichte des Maschinenbaues* (Berlin, 1899; 2nd edn., 1900).

<sup>9</sup> Abbott Payson Usher, *A History of Mechanical Inventions* (New York, 1929; revised edn., 1954).

<sup>10</sup> Charles Singer, *et al.*, editors, *A History of Technology* (5 vols., Oxford, 1954-58).

<sup>11</sup> William Barclay Parsons, *Engineers and Engineering in the Renaissance* (Baltimore, 1939). In the second volume of Singer's *History of Technology* a lift-lock system of Francesco di Giorgio is reproduced after Parsons and referred to an anonymous fifteenth century manuscript.

<sup>12</sup> Giovanni Canestrini, *Arte Militare Meccanica Medioevale* (Milan, n.d.).

<sup>13</sup> Umberto Forti, *Storia della Tecnica, dal Medioevo al Rinascimento* (Florence, 1957).

<sup>14</sup> Papini, *op. cit.*

<sup>15</sup> Friedrich Klemm, *Technik, eine Geschichte ihrer Probleme* (Freiburg and Munich, 1954).

book<sup>16</sup> appeared after the termination of this study of Francesco's contribution to the mechanical arts; the Francesco di Giorgio drawing on the book jacket is a sign of the importance the author attributes to the work of the Sienese engineer. In White's volume several of Francesco di Giorgio's projects in the *Trattato* and elsewhere are mentioned: an anti-scaling device for use on battlements, compound cranks and connecting rods, ball and chain governor, a saw-mill. Indeed, two of the ten plates that illustrate Professor White's work are reproductions from the *Trattato*. However, no historian of technology seems to have noticed the coincidences that form the subject of this essay.

It is beyond the scope of this study to give a detailed account of the machines in the Francesco codex, still less to evaluate their place in the history of technology. However, even a rapid examination of the drawings rewards the scholar with a surprising number of discoveries, for, although the *Trattato* was written around 1475, among the drawings we can observe many mechanical devices generally attributed to later epochs.

In these folios we can identify no less than 50 different types of flour and roller mills, including horizontal windmills, antedating Besson and Veranzio by a hundred years;<sup>17</sup> sawmills, pile-drivers, weight transporting machines, as well as all kinds of winches and cranes; roller-bearings and anti-friction devices; mechanical cars like those described by Dürer; a great number of pumps and water-lifting devices, including a machine similar to one designed by Wirz in 1746,<sup>18</sup> and an extremely interesting water or mud-lifting machine that must be characterized as the prototype of the centrifugal pump. Francesco not only designed polygonal fortifications earlier than either Dürer or Buonaiuto Lorini but described original war-machines, offensive and defensive, including the hydraulic recoil system for guns.<sup>19</sup> There are also devices for diving and swimming almost identical with those drawn by Leonardo da Vinci in his Manuscript B.

The question naturally arises as to why the Francesco di Giorgio manuscripts and apographa, containing such a wealth of information, remained unnoticed for so many centuries. The answer is that they

<sup>16</sup> Lynn White, Jr. *Medieval Technology and Social Change* (Oxford, 1962).

<sup>17</sup> According to Rex Wailes in Singer's *History of Technology* (Vol. II, p. 626), the earliest known illustration of a horizontal windmill is found in Besson, *Théâtre des Instrumens Mathématiques* (Lyons, 1579).

<sup>18</sup> Thomas Ewbank describes this machine in *A Descriptive and Historical Account of Hydraulic and other Machines for Raising Water* (New York, 1850), p. 363.

<sup>19</sup> Canestrini, *op. cit.*

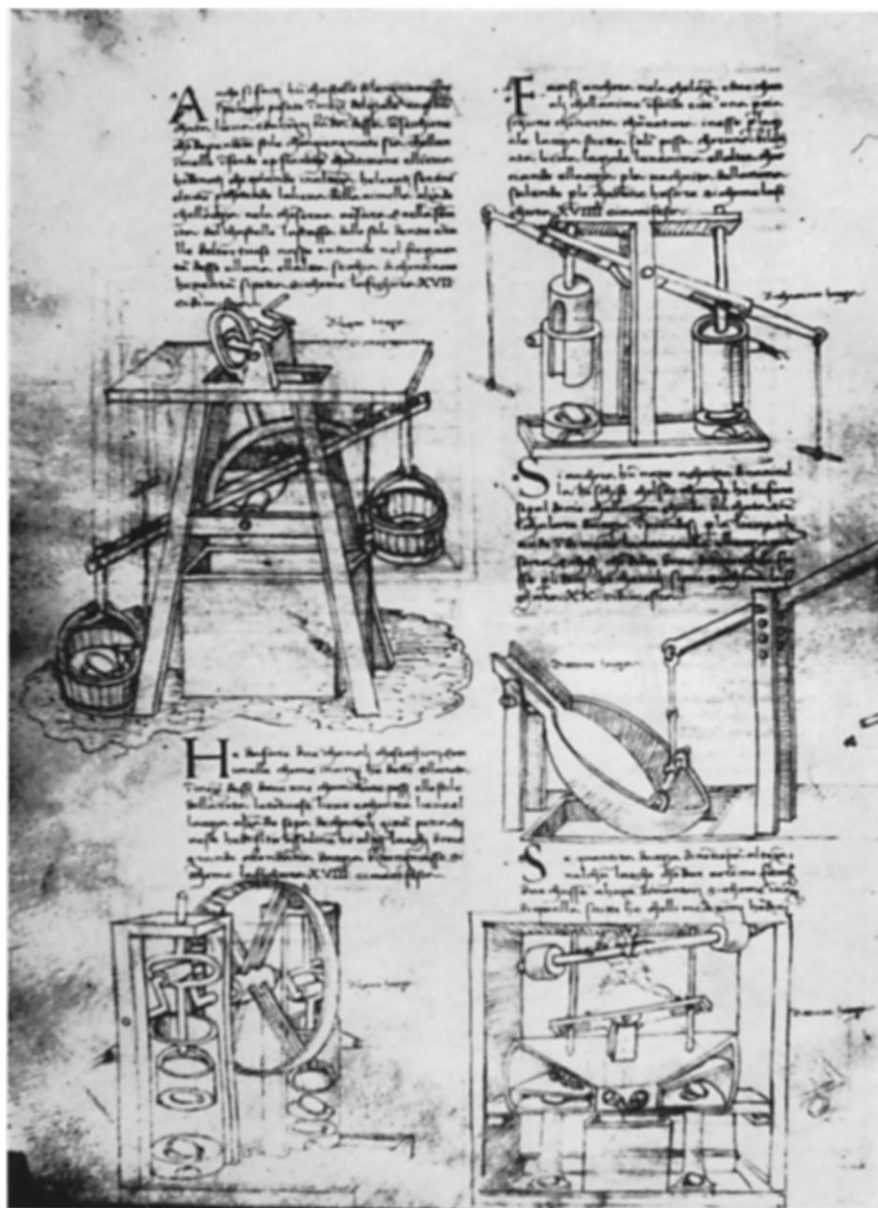


Fig. 1. Francesco di Giorgio Martini: A page from the Codex Saluzziano.



Fig. 2.  
Francesco di Giorgio:  
A roller mill  
driven by horizon-  
tal water-wheel.

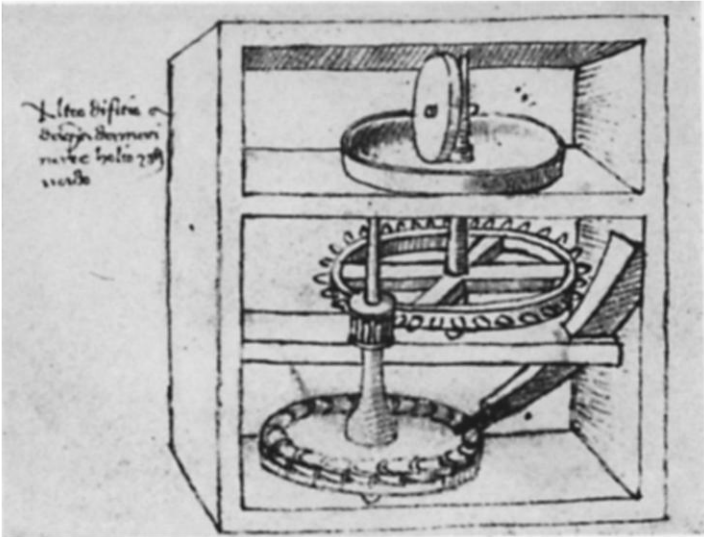
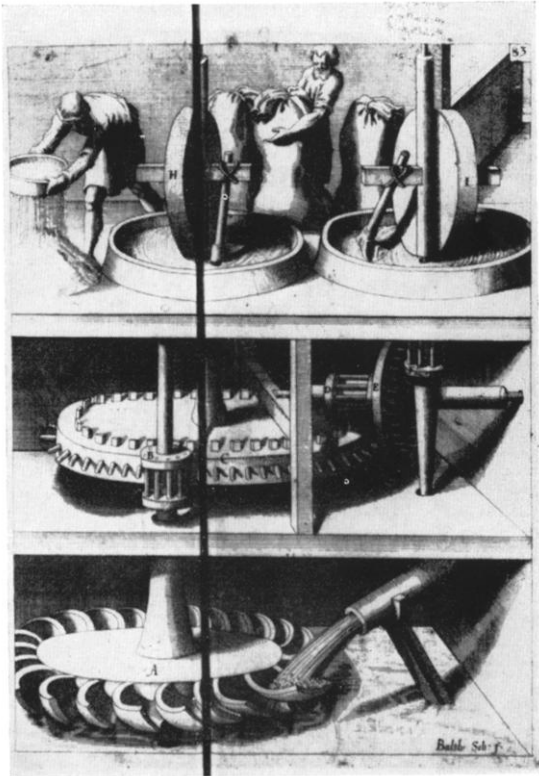


Fig. 2a. Strada: Plate 83.



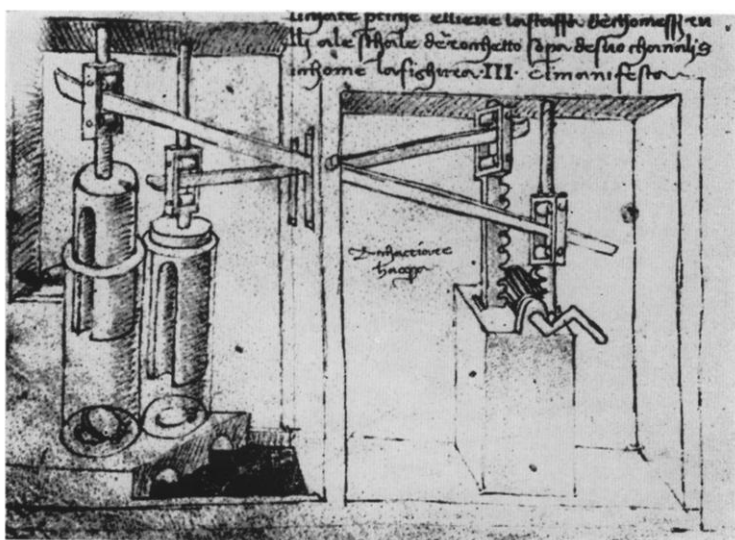


Fig. 3. Francesco di Giorgio: Double Pump.

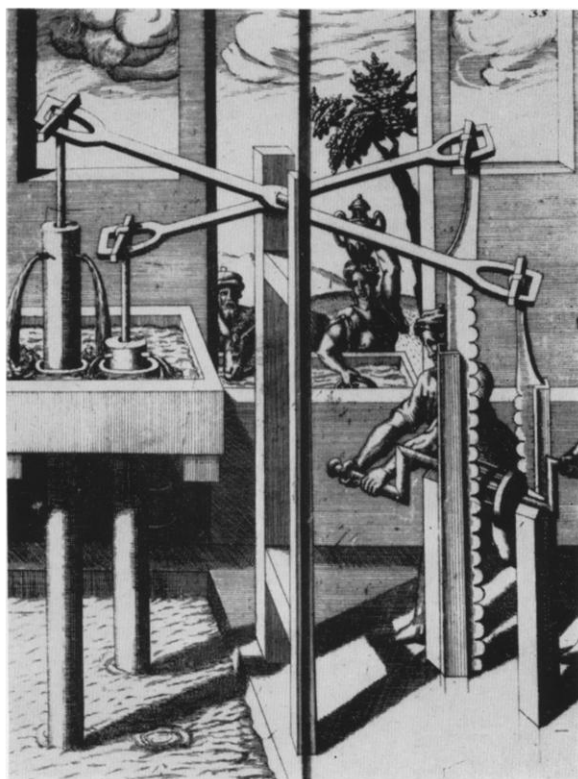


Fig. 3a. Strada: Plate 35.



Fig. 4.  
Francesco di Giorgio:  
Treadmill for a horse.

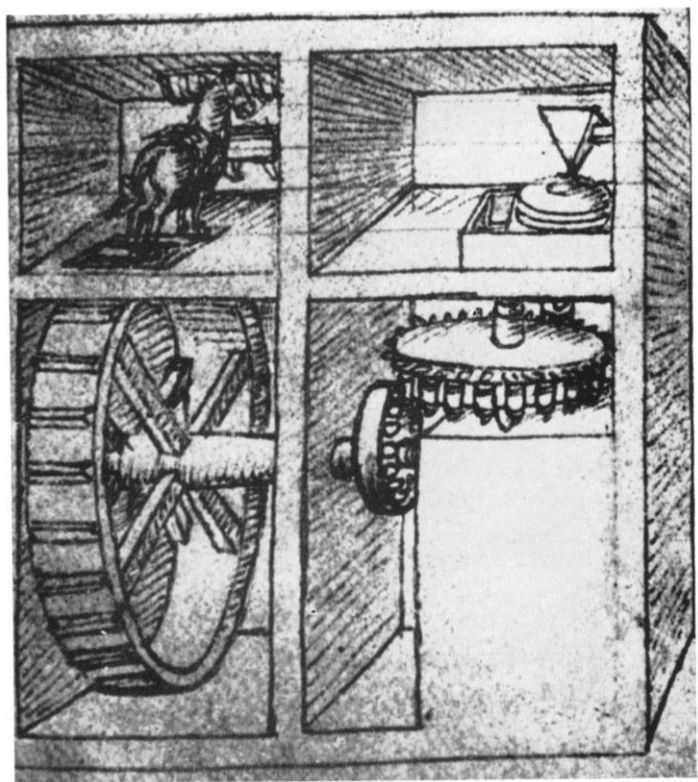
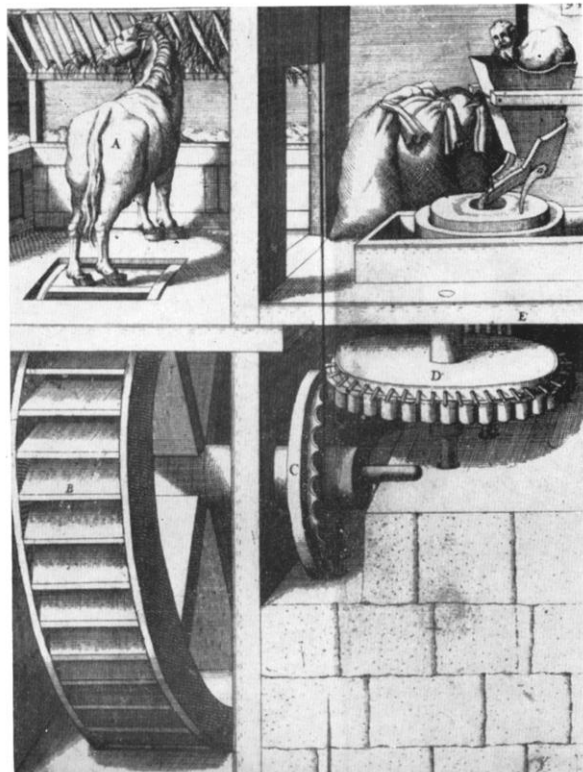


Fig. 4a.  
Strada: Plate 94.



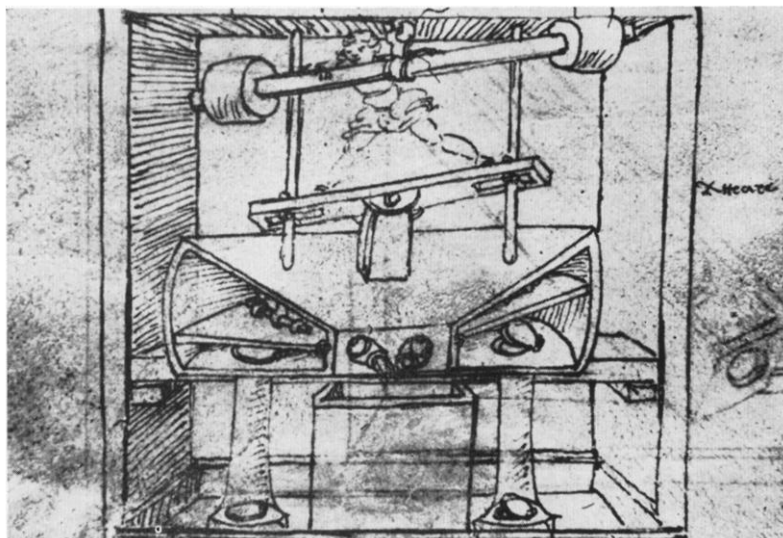


Fig. 5. Francesco di Giorgio: Double Pump with swinging piston.

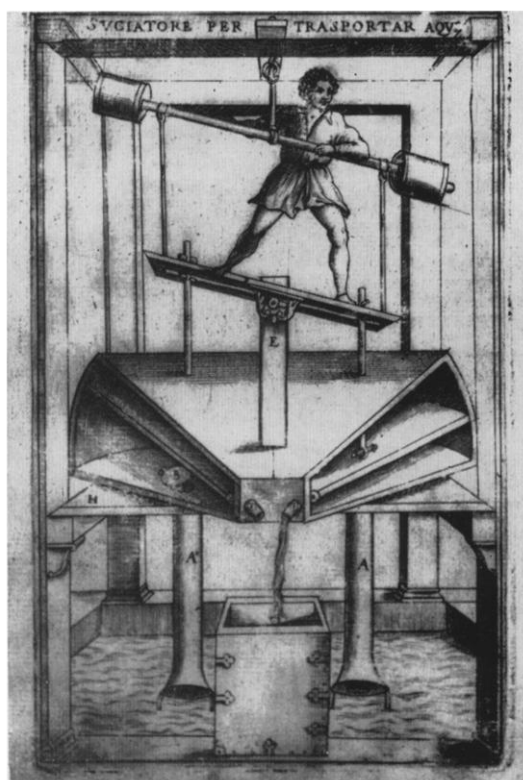


Fig. 5a. Zonca: Plate 100.

Fig. 6.  
 Leupold:  
*Theatrum Mach Hydr.*,  
 Vol. II, Plate 56, fig. 2.

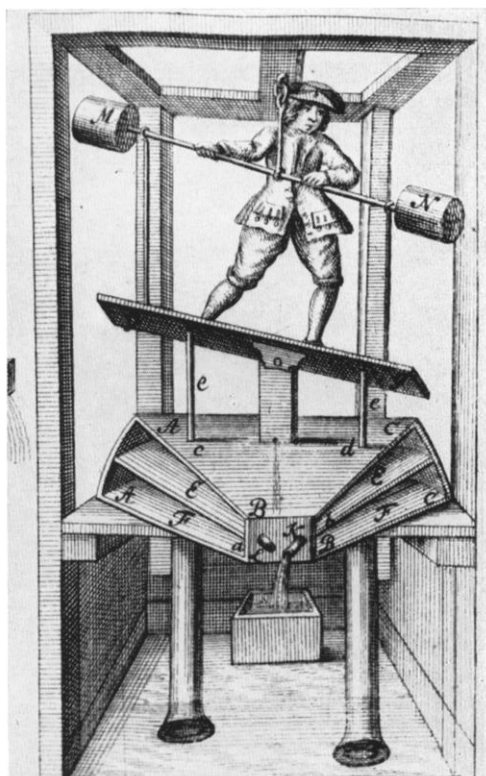
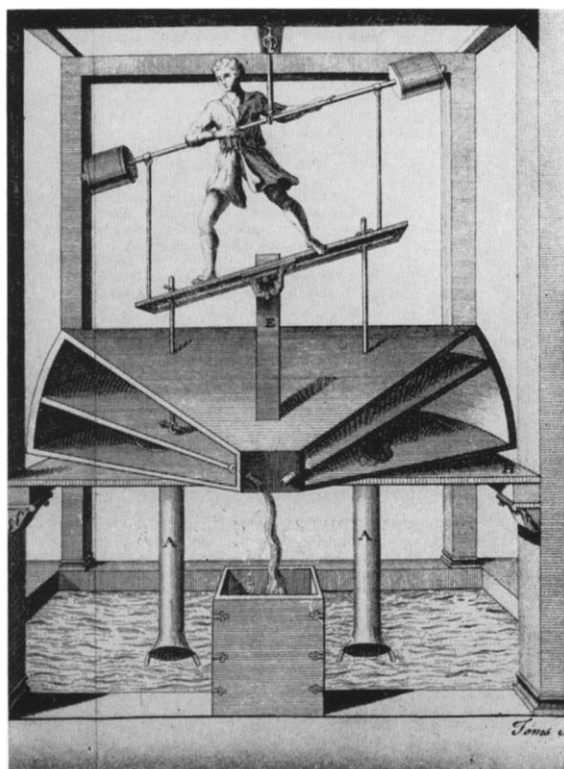


Fig. 7.  
 Switzer: Vol. II, Plate 15.



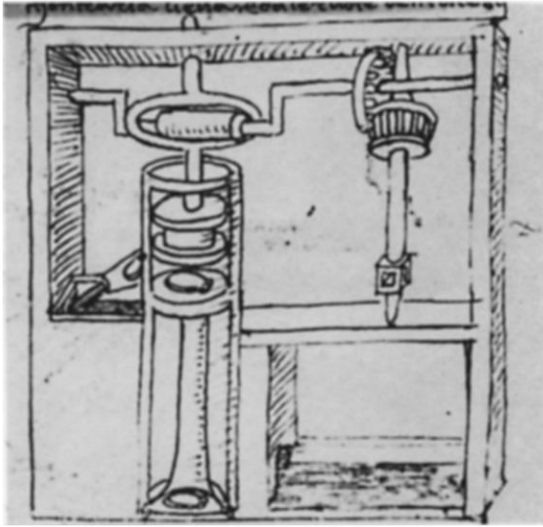


Fig. 8. Francesco di Giorgio:  
Single Suction Pump.

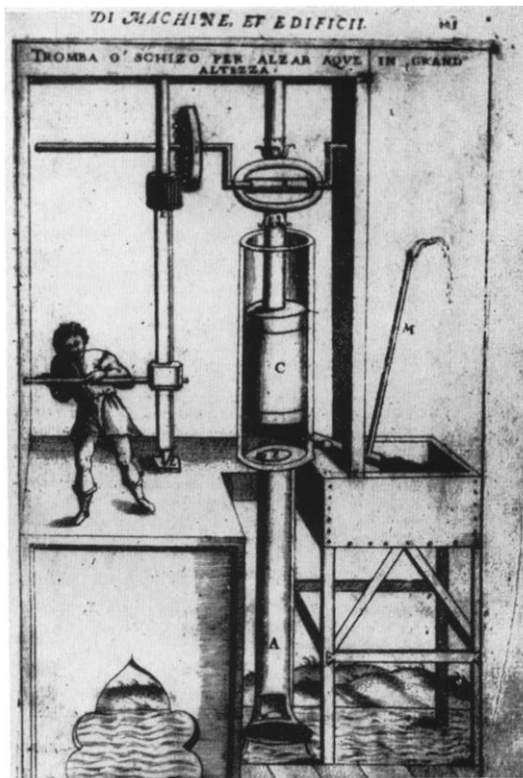


Fig. 8a. Zonca: Plate 103.



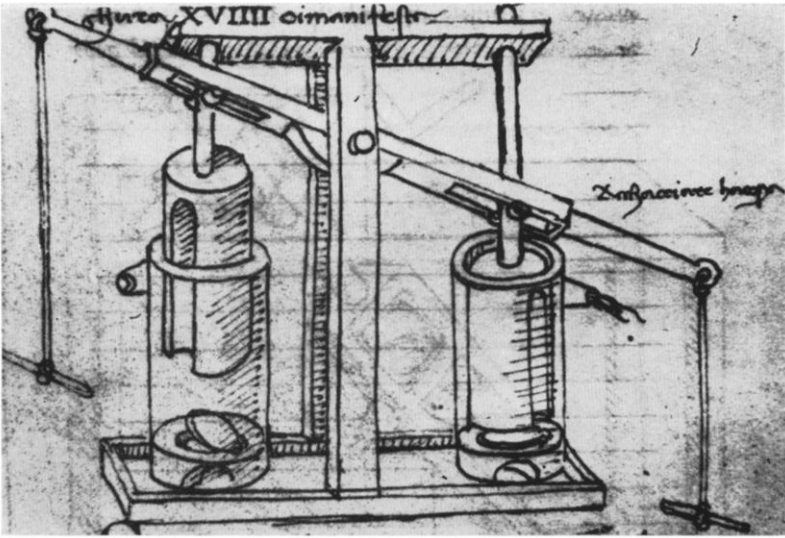


Fig. 9. Francesco di Giorgio: Double Suction Pump.

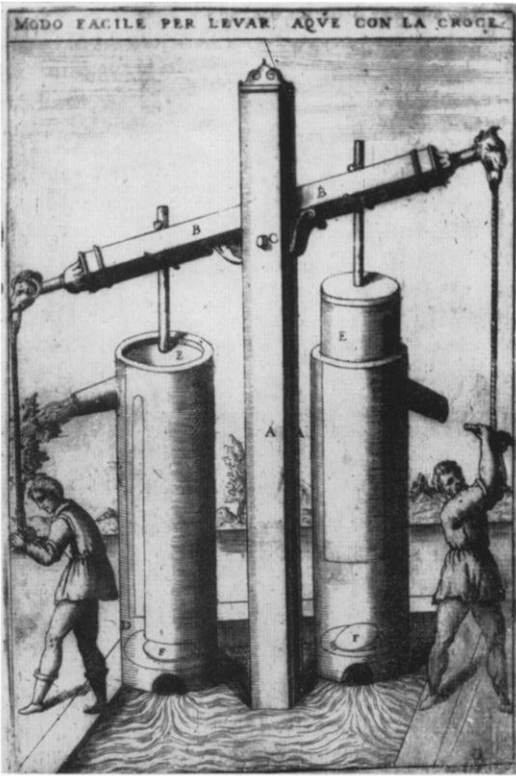


Fig. 9a. Zonca: Plate 105.



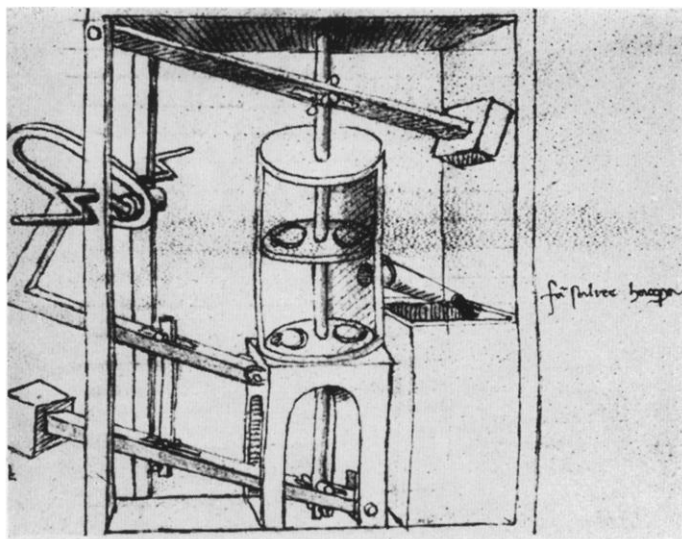


Fig. 10. Francesco di Giorgio: Single Suction Pump with counter-balance mechanism.

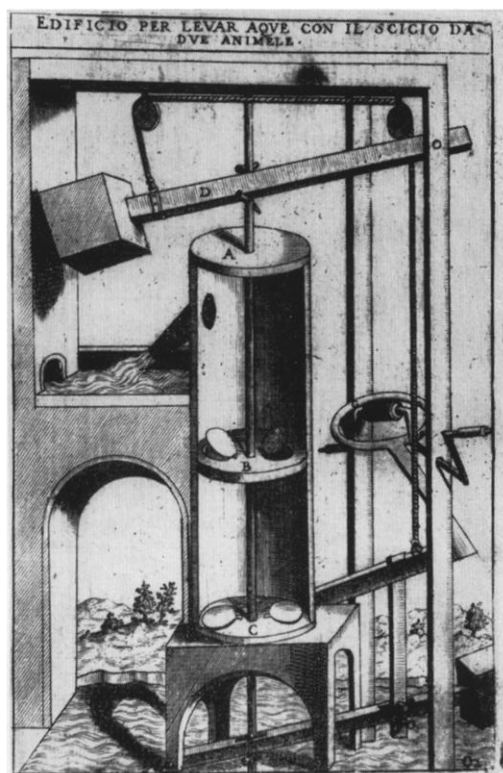


Fig. 10a. Zonca: Plate 109.

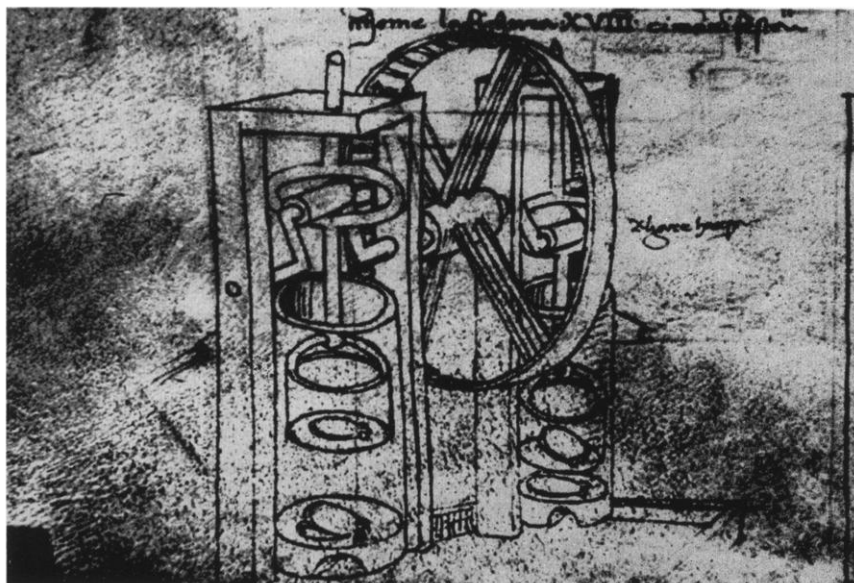


Fig. 11. Francesco di Giorgio: Double Suction Pump and Treadmill.

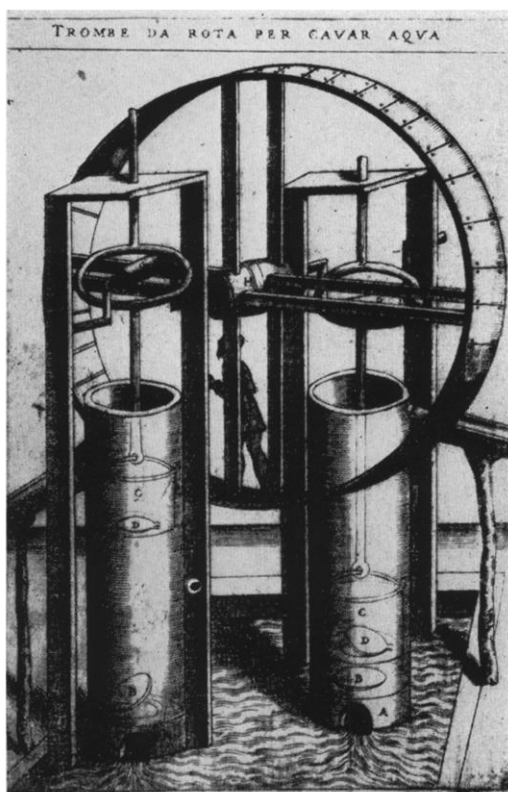


Fig. 11a. Zonca: Plate 110.

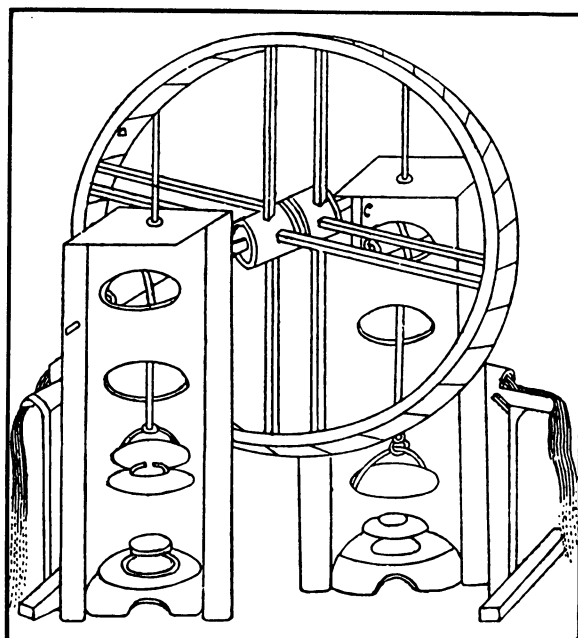


Fig. 11b. Chinese Encyclopedia, from Zonca, Plate 110.

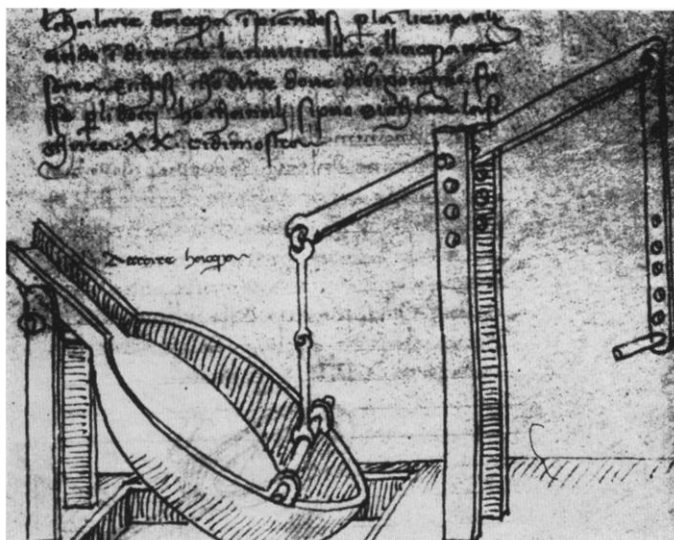


Fig. 12. Francesco di Giorgio: Water Scoop.

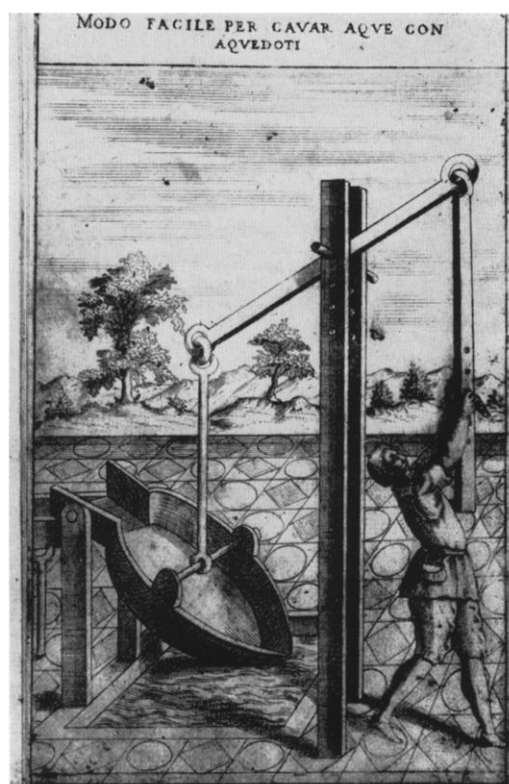


Fig. 12a. Zonca: Plate 112.



Fig. 13.  
Francesco di Giorgio:  
Flourmill. Driven by  
horizontal water-wheel  
and water recirculation  
that leads to a  
perpetuum mobile.

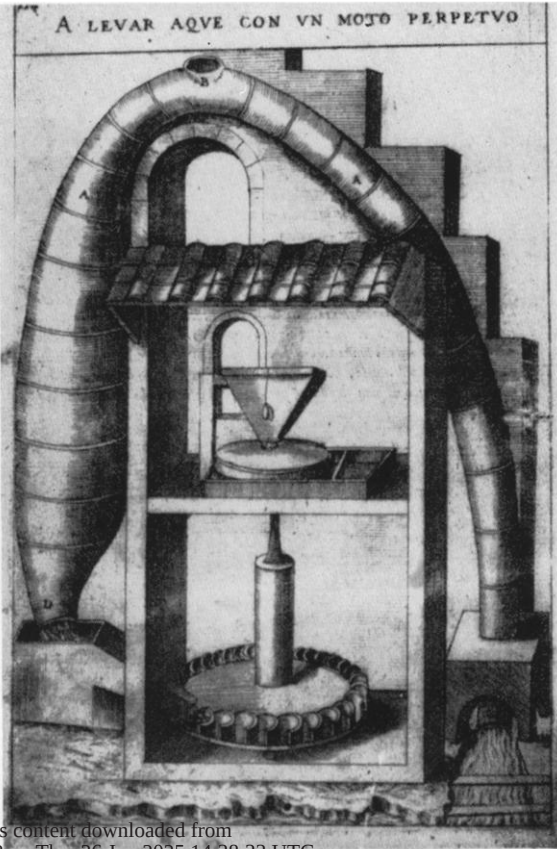


Fig. 13a. Zonca: Plate 117.

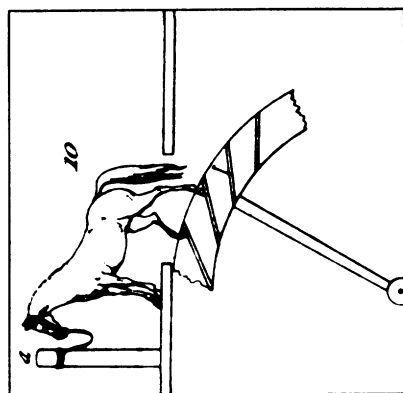


Fig. 14b: Borgnis: *Mecanique appliquée aux Arts*, Pl. 2. Horse on Treadmill after Strada, Plate 94 (Fig. 4a).

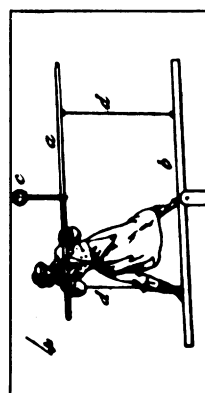


Fig. 14c. Borgnis. Swinging Mechanism (Pl. II), from Zonca, Plate 100 (Figure 5a).

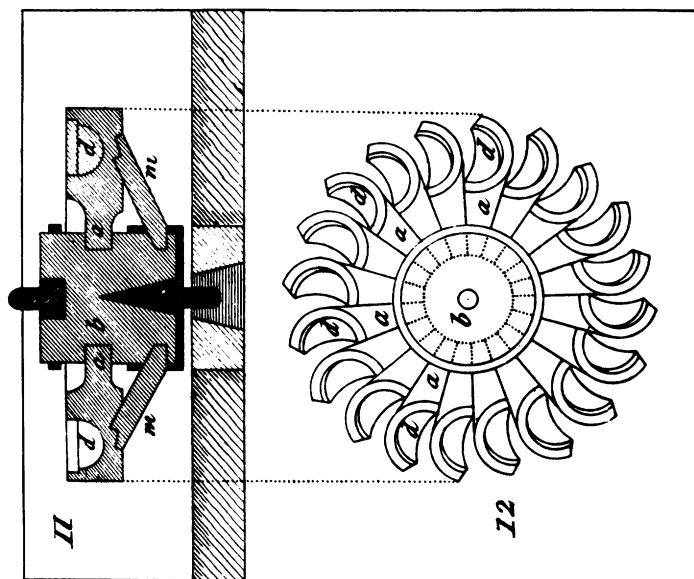


Fig. 14a. Borgnis: Construction of a scooped horizontal water-wheel. Cf. Strada, Plate 83 (Fig. 2a).

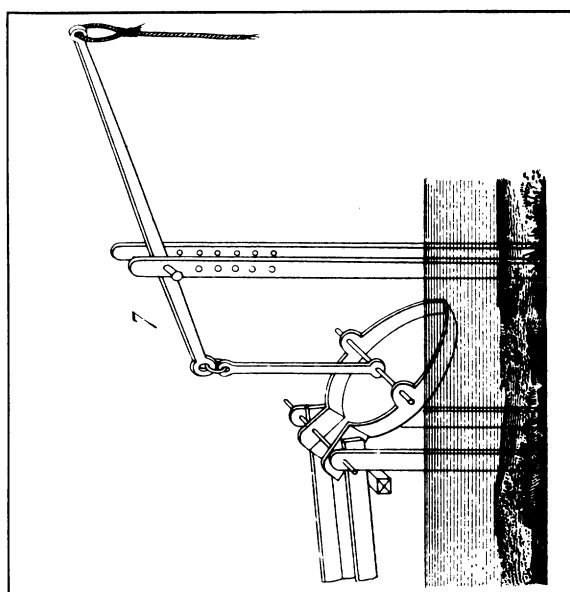


Fig. 14d. Borgnis: *Traité des Mech. hydrauliques*, Pl. 2. Water Scoop, like Zonca, Pl. 112 (Fig. 12a).



did not: The important ideas he describes were taken over without credit by writers of the sixteenth century. Some of the very authors credited with the transmission of the Leonardo heritage in the field of engineering—and following the suggestions of Duhem, Beck, and Usher we agree in naming the Italian school of engineers: Ramelli,<sup>20</sup> Besson,<sup>21</sup> Zonca,<sup>22</sup> Veranzio,<sup>23</sup> Castelli,<sup>24</sup> Strada,<sup>25</sup> etc.—found ample inspiration in the writings of Francesco di Giorgio.

The plundering must have begun during the author's lifetime, for Francesco himself laments it in the prologue to Book VII of the *Trattato*. His lucid and prophetic account proves beyond doubt that Francesco is indeed the author of the Laurenziana codex. On the paternity of the Saluzzo codex everybody is in agreement; the Laurenziana copy is anonymous and differs from the Saluzzo in several instances. Nevertheless, the part dedicated to machines is almost identical in both codices in text and figures. Since it is precisely this part which has been considered an arbitrary addition, it is important to quote Francesco's words wherein he definitely states that Book VII contains matters essential to complete what has been said about architecture in the first six books. He then reflects on the fate of inventors, and this, in view of the present study, has not a little pathos; it also proves, incidentally, that many of the machines and mechanisms described must have been devised by Francesco.

But let Francesco speak for himself in the prologue of Book VII:

In building it is necessary to move great weights, (for) without machines (*ingegno*) the force of man is of small avail; likewise, there is need to draw water in large quantity and convey it long distances. No less useful and necessary will be the construction of mills, and in some places where little water is available ingenuity must be used to help; in other places where there is no water at all, mills must be constructed that work with the wind or by some other means. Therefore, to conclude my work, in this last part I shall present examples from among the most common, potent, and useful of each of the said devices in spite of having several times decided not to disclose the nature of some of my machines and instruments. Such knowledge of them as I have has been acquired with great toil and at the sacrifice of my means of

<sup>20</sup> Agostino Ramelli *Le diverse et artificiose machine* . . . (Paris, 1588).

<sup>21</sup> James Besson, *Théâtre des Instrumens Mathématiques* (Lyon, 1579).

<sup>22</sup> Vittorio Zonca, *Novo Teatro di Machine et Edificii* (Padua, 1607).

<sup>23</sup> Fausto Veranzio, *Machinae Novae* . . . (Venice, 1615 [?]).

<sup>24</sup> Benedetto Castelli, *Della misura dell'acque correnti* (Rome, 1628).

<sup>25</sup> Jacob de Strada, *Künstlicher Abriss allerhand Wasser-Wind, Ross-und-Hand-mühlen* (Frankfurt, 1617-18).

livelihood, so I am reluctant to show them forth to all, for once an invention is made known not much of a secret is left. But even this would be a lesser evil if a greater did not follow. The worst is that ignoramuses adorn themselves with the labours of others and usurp the glory of an invention that is not theirs. For this reason the efforts of one who has true knowledge is oft retarded. If in all epochs this vice hath abounded, in our own it is more widespread than in any other.

Francesco then proceeds to tell of engineers who have taken advantage of his inventions but refrains from naming them "to avoid having my words attributed to passion."<sup>26</sup> He ends by saying: "Often such imitators are appreciated higher than the true craftsman, because no man is a prophet in his own country. However my countrymen (the Sienese) had not incurred this vice of ingratitude. On the contrary, in spite of my poor experience and insufficiency, they had sought to honor me and bestow on me great rewards. For this reason I have to exalt and praise my country. Etc."

We must not judge earlier writers by present-day standards in the use they made of the work of their predecessors. During the Middle Ages and the Renaissance, authors and sources were cited only in case of refutation; technical drawings and sketches came down from time immemorial suffering additions and modifications. It is also highly probable that the mechanical projects shown in Francesco's *Treatise* were, in part, known at the time.<sup>27</sup> To render true justice it would be necessary to examine carefully and compare the scanty remains of technological skills hidden in the few surviving manuscripts left by engineers of the first half of the fifteenth century. It is certain that some of the ideas illustrated by Francesco are older,<sup>28</sup> but there is no question as to the artistic originality of his rendering.

Most of the Francesco di Giorgio machines shown in the *Trattato* differ from those of his predecessors in the singular composition of the drawing. For example, Fig. 1, a reproduction of a complete page of the *Trattato* manuscript showing how the machine drawings appear

<sup>26</sup> Promis is sure that Francesco di Giorgio is referring to a Florentine engineer and considers as possible one of the following: Filarete, Bernardo Fiorentino, or perhaps Giuliano da San Gallo, but as most likely Baccio Pontelli. He emphatically excludes Leonardo da Vinci from the list.

<sup>27</sup> Francesco quite honestly avows that in his treatise he has "extracted many conclusions from several books" (Promis, *Trattato*, p. 128).

<sup>28</sup> In particular it would be rewarding to compare and study the manuscripts of Jacopo Mariano, called Il Taccola, to establish former derivations. Cf. Lynn Thorndike, "Marianus Jacobus Taccola," *Archives internationales d'histoire des sciences*, VIII (1955), p. 7.

on the folios of the codex, displays Francesco's characteristic originality in confining most of the mechanisms depicted to a frame that makes them appear, to use a modern expression, like a packaged unit. This arrangement is so characteristic that it could be used to trace the influence of Francesco in the machine books of the sixteenth and seventeenth centuries. Further, the drawings are of such technical excellence as only to be surpassed by Leonardo himself. This same artistic quality enables us in many instances to distinguish between legitimate influence and shameless plagiarism.

Disregarding all those cases where a derivation has been cunningly concealed by changing irrelevant details (particularly to be observed in the projects of Strada) or where an indirect influence may come into question (as in the case of Ramelli and Veranzio), I shall limit myself to pointing out just a few of the most evident borrowings. First we shall compare Francesco's drawings with some of the machines described and engraved in Strada's famous book. Jacob di Strada of Rosberg (ca. 1523-1588) was *Antiquarium Commissarium Bellicum* to three Emperors: Ferdinand I, Maximilian II, and Rudolph II. His book, published posthumously, was edited by his nephew, Octavius Strada, and bears the date 1617-18. A French edition appeared at the same time; a second edition was issued in 1629. In 1661 almost all the plates were reproduced in Böckler's *Theatrum Machinarum Novum*.<sup>29</sup> This latter book enjoyed great popularity and went through several editions, the last in 1703; its predecessor is given only the most perfunctory acknowledgment in the foreword.

Among the Strada plates deriving from the Francesco codex I shall select a few where the influence is most strikingly manifest. In Figures 2 and 2a, we can compare Francesco's original with Strada's rendering. This drawing of a roller-mill is particularly interesting because it is moved by a geared horizontal water-wheel with scoops that recall those of the Pelton wheel. An unusual feature is the confining of the stream to a narrow tube, the jet effect making the wheel more efficient. Leonardo da Vinci, in *Cod. Atl. 2 recto a* and *355 recto b*, sketches a similar arrangement. Strada merely adds human figures and a second roller—technically an unsound proposal.

Figures 3 and 3a show a double pump. The anti-friction device employed by Francesco has not been understood by Strada, causing his project to be technically inferior, but the additions of the human figures and other decorations hardly disguise the derivation.

<sup>29</sup> Georgius Andreas Böckler, *Theatrum Machinarum Novum* (Nürnberg, 1661; other German editions, 1662, 1673, 1686, 1703; Latin editions: Col. Agripp., 1662; Norinbergae, 1686).

In Figure 4 and its counterpart in Strada, Figure 4a, we discover a well-known design which has appeared in many books on mechanical technology: a flour-mill driven by a horse on a treadwheel. From Strada the drawing found its way to Böckler and other *Theatra*, finally appearing in Leupold's treatise,<sup>30</sup> where it is mechanically examined. Leupold in his books reproduced and analyzed several of the Francesco di Giorgio projects, using the intermediary sources of Strada, Zonca, Böckler, and Zeising. Quite interesting is Beck's comment: "Tread-wheel moved by the hind legs of a horse as shown in Figure 813 driving a mill. We found an identical mill still in use in Holland about 30 years ago (that is, in 1870). The way in which the teeth of the angle wheels in this plate—and in several other plates of Strada—mesh together is peculiar. Mounted on the shaft of the tread-wheel there is a wooden wheel with a metal toothed crown with semi-circular cut-outs and the wooden teeth that mesh with it are half-cylindrical."<sup>31</sup> This is an interesting example of the survival of a mechanical invention. The mill seen in Holland is, however, at least a hundred years older than Beck supposed, dating back to the fifteenth century. Strada must have been a poor mechanic, as was pointed out in an earlier case. He mistook the so-modern looking roller-bearings of Francesco for a round-toothed angle-wheel to which he felt obliged to apply the semi-circular cut-out cogs of the driving wheel. This explains the problem that Beck so acutely noticed sixty years ago.

Many other Strada machines manifestly derive from the same source, a hundred years older. Strada tried, not always successfully, to cover his appropriations not only by introducing scenery and figures but also by technical variations that compel us to be cautious in our conclusions.

The second author, Vittorio Zonca (1568-1602), was less careful. His book enjoyed great popularity and went through four editions, the last in 1656. In Beck's biography of Zonca we read, "Publication of so many editions demonstrates how highly the book was esteemed in its day and the capacity of its author."<sup>32</sup>

Just as the Strada engravings found ready appreciation in Böckler, so many of Zonca's were incorporated without credit in another

<sup>30</sup> Jacob Leupold, *Theatrum Machinarum Hydraulicarum* (2 vols., Leipzig, 1724); *Theatrum Machinarum Generale*, *Theatrum Machinarum Hydrotechnicarum* (Leipzig, 1724).

<sup>31</sup> Beck, *op. cit.*, pp. 535, 537.

<sup>32</sup> *Ibid.*, p. 294.



popular compendium, the *Theatrum Machinarum* of Heinrich Zeising.<sup>33</sup> This book was republished a number of times from its first edition in 1607-14 to its last in 1708. Comparing the last seven figures in Zonca's treatise, all but one of them illustrating water-lifting devices, with the corresponding Francesco di Giorgio drawings (Figures 5 to 12a), we find that Zonca's plates are straightforward reproductions of Francesco's drawings.

Figure 5 is another classic in the history of mechanical inventions, a double pump with a swinging piston. Without discussing the merits of this contrivance (Leupold does), we can compare it with the Zonca plate (Figure 5a). The Francesco drawings pirated by Zonca must have had some kind of fascination for the authors of books on machines in the seventeenth and eighteenth centuries. They were adopted by Zeising (Plates 12-18 of Part II) and from there passed on to Leupold. In England they became known by way of an important treatise by Switzer<sup>34</sup> that described some of the Zonca water-lifting devices (Pls. 13-15 of Volume II), attributing them to "an Italian author of great repute in machines." Figure 6 shows the swinging piston pump as it is in Leupold, who has given the worker a contemporary suit of clothes, while Figure 7 is Switzer's version.

A few words are needed to illustrate the Francesco drawings used by Zonca for the last copper-plate of his book (Cf. Figures 13 and 13a). It represents a flour-mill to be set in motion by a horizontal water-wheel similar to the one copied by Strada shown in Figure 2. This wheel is for use where there is no running water; circulation of the water is obtained by an artifice that leads to a *perpetuum mobile*. It is surprising that Zonca praises "his object" as *una inventione non mai piu usata*. Several other mills among the Francesco drawings are based on similar hydraulic paradoxes that Francesco calls *molini per acqua morta*. Perpetual motion was a mirage that had deluded men since time immemorial; in justice to the mechanicians of the fifteenth, sixteenth, and seventeenth centuries it should be said that the idea was not considered absurd till very recent times. In fact, it was not until 1775 that the French Academy of Sciences resolved to reject schemes for perpetual motion machines in the future.<sup>35</sup>

Francesco's images did not stop there; they were passed on in hand-books on mechanics till the beginning of the nineteenth century—a

<sup>33</sup> Heinrich Zeising, *Theatrum Machinarum, so in sechs Theil bestehend* . . . (Leipzig, 1607-14).

<sup>34</sup> Stephan Switzer, *Introduction to a General System of Hydrostaticks and Hydraulicks* (2 vols., London, 1729).

<sup>35</sup> Henry Dircks, *Perpetuum Mobile* (2 vols., London, 1861-70).

curious cavalcade that started around 1470. This can be seen very clearly if we take a look at an important treatise of the early nineteenth century: J. A. Borgnis, *Traité complet de mécanique*. . . .<sup>36</sup> Looking at some of the basic mechanisms reproduced, we observe that they still bear the characteristic traits and style of the prototypes in the Francesco di Giorgio manuscripts (Figures 14a, 14b, 14c, and 14d). This is an amazing example of the survival and transmission of technical imagery. Of course, technology had made tremendous advances after the ingenious yet often naive contrivances of Francesco and his imitators. Nevertheless, as in the realm of the visual arts (and machine drawing is indeed an art subject to the same trends of evolution as art generally), the past inescapably molds the present and the future. The traditional delineation of machines that for centuries duplicated the patterns established by Francesco di Giorgio could not be abandoned when the mechanical contrivances they referred to had become obsolete; the older models had become technological symbols, as can be seen in several treatises of the eighteenth and nineteenth centuries.

The technological tradition that started with Francesco di Giorgio found its way, and quite early, to the Far East. Sabbatinus de Ursis (1575-1620) was assistant to the famous Jesuit missionary Matteo Ricci (1562-1610), who opened China to Christianity, and later became Superior to the mission. Among other important contributions, de Ursis (known in China as *Hsiung San-Pa*) redacted in 1612 an illustrated book with the title *Machines of the West*. Here contemporary European technology was presented to the Chinese technicians. For his compendium de Ursis utilized *inter alia* the books of Besson, Veranzio, Ramelli, and Zonca. Unfortunately, the Chinese draftsmen, unacquainted with Western technological perspective, could not copy the machine drawings properly (See Fig. 11 b). In 1726 this book was incorporated in the great Chinese Encyclopedia (5020 volumes). When European scholars studied this monumental work between 1865-88 they were greatly impressed by the technological knowledge they found therein. Only later was it realized that its sources were European.<sup>37</sup>

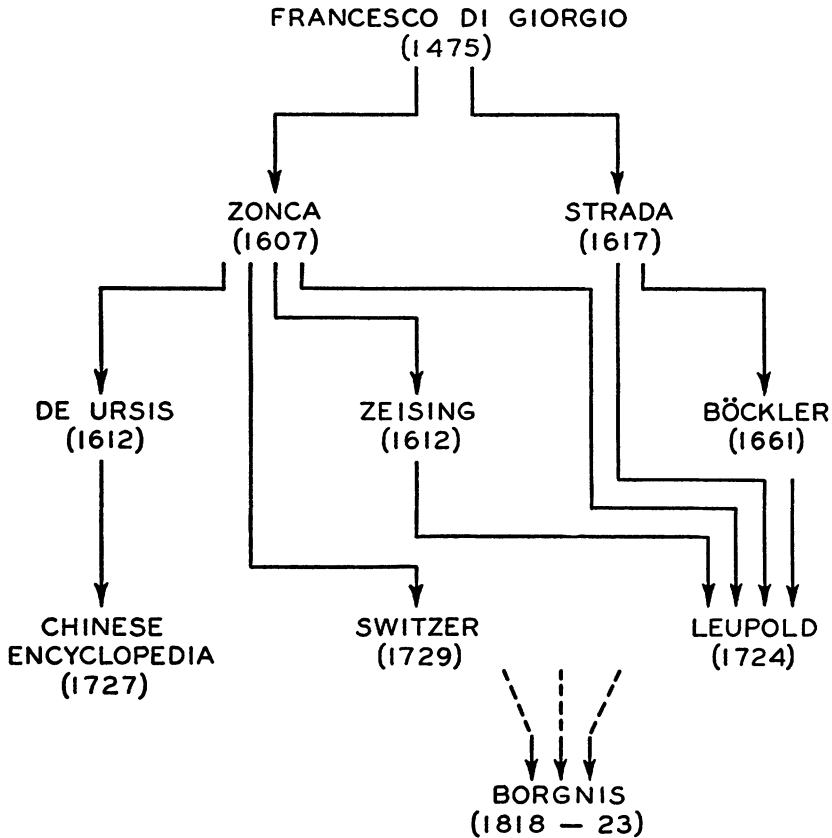
In the following diagram I have tried to show the main steps by which the Francesco di Giorgio machine projects were transmitted and incorporated in the fast developing technology of Europe and the Far East. Of course it is only a schematic representation and in some

<sup>36</sup> J. A. Borgnis, *Traité complet de Mécanique appliquée aux Arts* . . . (10 vols., Paris, 1818-23).

<sup>37</sup> F. M. Feldhaus, *Die Maschine im Leben der Völker* (Basel and Stuttgart, 1954), p. 58.

cases cross-paths are evident (e.g., Böckler took over plates from either Zonca or Zeising). The derivations are more numerous than stated here, but it would be a monumental task to trace them all in the voluminous technical literature of the seventeenth and eighteenth

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centuries. Francesco di Giorgio's design for a pile-driver provides an interesting case. The principle must be considered original with Francesco but it appears in a book published in Bologna in 1678 by Alessandro Capra.<sup>38</sup> In his introduction Capra tells us that this is "a new and beautiful invention that has been used by me in building forti-

<sup>38</sup> Alessandro Capra, *La Nuova Architettura Famigliare* (Bologna, 1678).

fications." It may be added that Francesco's project is mechanically superior.

It should be noted that the methods followed in this preliminary study of Francesco di Giorgio's contributions to the history of mechanical inventions are analogous to those used in a completely different field, the history of art. Such scholars as Arnold von Salis,<sup>39</sup> Erwin Panofsky,<sup>40</sup> Sir Kenneth Clark,<sup>41</sup> and K. E. Maison<sup>42</sup> have published extraordinarily interesting studies on the survival of antique imagery through the ages and on the compelling forces that lead artists of the first excellence to use the formal motifs of their predecessors. What holds true in the history of art would seem also to be true of this particular aspect of the history of technology.\*

<sup>39</sup> Arnold von Salis, *Antike u. Renaissance. Über Nachleben u. Weiterwirken des Alten in der neueren Kunst* (Erlenbach and Zürich, 1947).

<sup>40</sup> Erwin Panofsky, *Meaning in the Visual Arts* (Garden City, N. Y., 1957).

<sup>41</sup> Kenneth Clark, *The Nude* (London, 1956).

<sup>42</sup> K. E. Maison, *Themes and Variations* (London, 1960).

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The machine drawings here reproduced are all from the Turin Saluzziano Codex because the damage suffered by the Laurenziana Ashb. 361 rendered it less suitable for reproduction. However, the Laurenziana Codex, or a copy of it very carefully executed, was used by Vittorio Zonca. There are details in this codex which do not appear in the Saluzziano but which are incorporated in Zonca's rendering. A further proof that a copyist is at work is the fact that in Zonca's book all but one of the figures appear in mirror image of the original manuscript drawings in the Laurenziana codex, a well-known occurrence when careless copyists transfer a drawing to a copper-plate.